

=== BC-PC-TSP source=city_rewards.csv nodes=115474 budget=3100000 runs=10
W=29178 ===

Algorithm	Rewards	Dist	Remaining	Time(ms)
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— Run 1/10 (start city=67911) —

Greedy1	108365	3099998.43	1.57	65
Greedy2	2830577	3100000.00	0.00	929256
PMARL	2805524	3100000.00	0.00	832940
AntQ	2474937	3099999.71	0.29	250951

— Run 2/10 (start city=91691) —

Greedy1	110194	3099998.25	1.75	15
Greedy2	2393505	3099999.93	0.07	818413
PMARL	2887644	3099999.98	0.02	847091
AntQ	2655793	3099999.87	0.13	358687

— Run 3/10 (start city=91691) —

Greedy1	110194	3099998.25	1.75	15
Greedy2	2393505	3099999.93	0.07	849293
PMARL	2887644	3099999.98	0.02	843515
AntQ	2655793	3099999.87	0.13	352744

— Run 4/10 (start city=96072) —

Greedy1	108895	3099995.63	4.37	14
Greedy2	2358457	3099999.38	0.62	873429
PMARL	2861546	3099997.70	2.30	839456
AntQ	2698912	3100000.00	0.00	95490

— Run 5/10 (start city=92478) —

Greedy1	110259	3099999.99	0.01	14
Greedy2	3137226	3100000.00	0.00	1108251
PMARL	2866416	3099999.94	0.06	864990
AntQ	2905450	3099971.13	28.87	264486

— Run 6/10 (start city=75263) —

Greedy1	109991	3099985.44	14.56	14
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Greedy2	2924625	3100000.00	0.00	1013475
PMARL	2870293	3099999.87	0.13	856357
AntQ	2951329	3100000.00	0.00	107058

— Run 7/10 (start city=88303) —

Greedy1	110331	3099995.07	4.93	14
Greedy2	3150021	3100000.00	0.00	1104618
PMARL	2885589	3099999.90	0.10	869472
AntQ	2537751	3099999.96	0.04	92895

— Run 8/10 (start city=71706) —

Greedy1	109885	3099989.97	10.03	14
Greedy2	3016330	3100000.00	0.00	1018072
PMARL	2869744	3099999.85	0.15	865763
AntQ	2801897	3100000.00	0.00	73512

— Run 9/10 (start city=81585) —

Greedy1	109881	3099997.14	2.86	14
Greedy2	2855863	3099999.96	0.04	1015035
PMARL	2888806	3099999.99	0.01	869429
AntQ	2721833	3099999.99	0.01	86847

— Run 10/10 (start city=67911) —

Greedy1	108365	3099998.43	1.57	72
Greedy2	2830577	3100000.00	0.00	922498
PMARL	2805524	3100000.00	0.00	828706
AntQ	2474937	3099999.71	0.29	250942

1. Rewards

Algorithm	Mean	95% CI (t-distribution)	95% CI (z-distribution)
Greedy1	109,636.00	[109,075.67, 110,196.33] (Mean 560.33)	[109,150.52, 110,121.48] (Mean 485.48)

Greedy2	2,789,068.60	[2,572,116.53, 3,006,020.67] (Mean 216,952.07)	[2,601,098.35, 2,977,038.85] (Mean 187,970.25)
PMARL	2,862,873.00	[2,840,113.43, 2,885,632.57] (Mean 22,759.57)	[2,843,153.80, 2,882,592.20] (Mean 19,719.20)
AntQ	2,687,863.20	[2,569,521.70, 2,806,204.70] (Mean 118,341.50)	[2,585,330.51, 2,790,395.89] (Mean 102,532.69)

2. Execution Time (ms)

Algorithm	Mean	95% CI (t-distribution)	95% CI (z-distribution)
Greedy1	25.10	[8.69, 41.51] (Mean 16.41)	[10.88, 39.32] (Mean 14.22)

Greedy2	965,234.00	[892,114.78, 1,038,353.22] (Mean 73,119.22)	[901,882.51, 1028,585.49] (Mean 63,351.49)
PMARL	851,771.90	[840,728.18, 862,815.62] (Mean 11,043.72)	[842,203.47, 861,340.33] (Mean 9,568.43)
AntQ	193,361.20	[111,708.80, 275,013.60] (Mean 81,652.40)	[122,616.44, 264,105.96] (Mean 70,744.76)